

CBCS SCHEME - Make-Up Exam

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BCS303

Third Semester B.E./B.Tech. Degree Examination, June/July 2025 Operating Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C								
Q.1	a.	With neat diagram, Explain abstract view of the components of a computer system.	6	L3	CO1								
	b.	With neat diagram, explain virtual, non virtual and VM ware architecture.	8	L3	CO1								
	c.	Explain with neat diagram Dual-Mode operation.	6	L3	CO1								
OR													
Q.2	a.	Explain types of system calls provided by operating system.	6	L3	CO1								
	b.	Explain with neat diagram simple structure of MS-DOS layer structure and unix system structure.	8	L3	CO1								
	c.	Explain operating-system services.	6	L3	CO1								
Module – 2													
Q.3	a.	Explain process state with diagram.	5	L2	CO2								
	b.	What do you mean by interprocess communication? Explain two model of interprocess communication.	9	L2	CO2								
	c.	What are three types of multithreading models? Explain.	6	L2	CO2								
OR													
Q.4	a.	What do you mean by thread libraries? Discuss threading issues.	5	L2	CO2								
	b.	Consider the following set of four process with length of CPU burst given in MS: <table border="1"><thead><tr><th>Process</th><th>Burst time</th></tr></thead><tbody><tr><td>P₁</td><td>24</td></tr><tr><td>P₂</td><td>3</td></tr><tr><td>P₃</td><td>3</td></tr></tbody></table> Compute the waiting time and average turn around time for the above process using FCFS scheduling algorithm.	Process	Burst time	P ₁	24	P ₂	3	P ₃	3	10	L2	CO2
Process	Burst time												
P ₁	24												
P ₂	3												
P ₃	3												
	c.	With diagram explain SMT architecture.	5	L2	CO2								
Module – 3													
Q.5	a.	What is critical section problem? Explain Peterson’s solution and synchronization hardware solution for critical section problem.	10	L3	CO3								
	b.	Write a code for readers-writers process.	6	L3	CO3								
	c.	Discuss the structure of philosopher.	4	L3	CO3								

1 of 2

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OR																																																																											
Q.6	a.	What are the four necessary condition for deadlock occurrence?	4	L3	CO3																																																																						
	b.	Explain with neat diagram resource-allocation graph for deadlock avoidance.	6	L3	CO3																																																																						
	c.	Consider the following snapshot of the system: <table border="1"><thead><tr><th rowspan="2">Process</th><th colspan="3">Allocation</th><th colspan="3">Max</th><th colspan="3">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr></thead><tbody><tr><td>P₀</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P₁</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></tbody></table> Determine whether the system is safe using Banker's algorithm. If the request for P ₁ arrives for (1, 0, 2) can the request be granted immediately.	Process	Allocation			Max			Available			A	B	C	A	B	C	A	B	C	P ₀	0	1	0	7	5	3	3	3	2	P ₁	2	0	0	3	2	2				P ₂	3	0	2	9	0	2				P ₃	2	1	1	2	2	2				P ₄	0	0	2	4	3	3				10	L3	CO4	
Process	Allocation			Max			Available																																																																				
	A	B	C	A	B	C	A	B	C																																																																		
P ₀	0	1	0	7	5	3	3	3	2																																																																		
P ₁	2	0	0	3	2	2																																																																					
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P ₄	0	0	2	4	3	3																																																																					
Module – 4																																																																											
Q.7	a.	Explain the following with respect to dynamic storage allocation: i) First fit ii) Best fit iii) Worst fit	3	L3	CO4																																																																						
	b.	What is Paging? Explain with neat diagram paging hardware and paging model of logical and physical memory.	10	L3	CO4																																																																						
	c.	With neat diagram, explain segmentation hardware.	7	L3	CO4																																																																						
OR																																																																											
Q.8	a.	With neat diagram, explain demand paging system.	6	L3	CO4																																																																						
	b.	Consider the page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 for a memory with 3 frames. Determine the number of page faults using FiFO, optimal and LRU replacement algorithms. Which algorithm is most efficient?	9	L3	CO1																																																																						
	c.	Write a note on copy_on_write technique.	5	L3	CO1																																																																						
Module – 5																																																																											
Q.9	a.	Explain bit vector and linked free-space list on disc.	6	L3	CO5 CO6																																																																						
	b.	Explain with neat diagram contiguous allocation and indexed allocation.	8	L3	CO5 CO6																																																																						
	c.	With neat diagram single level and two level directory structure.	6	L3	CO5 CO6																																																																						
OR																																																																											
Q.10	a.	Discuss network attached storage.	5	L3	CO5 CO6																																																																						
	b.	A disk drive has 200 cylinders 0 to 199. Head starts at 53 to serve the request queue: 98, 183, 37, 122, 14, 124, 65, 67. Draw disk head schedule diagram and explain for FCFS, SSTF, C-SCAN and C-LOCK.	10	L3	CO5 CO6																																																																						
	c.	Explain the concept of access matrix.	5	L3	CO5 CO6																																																																						
